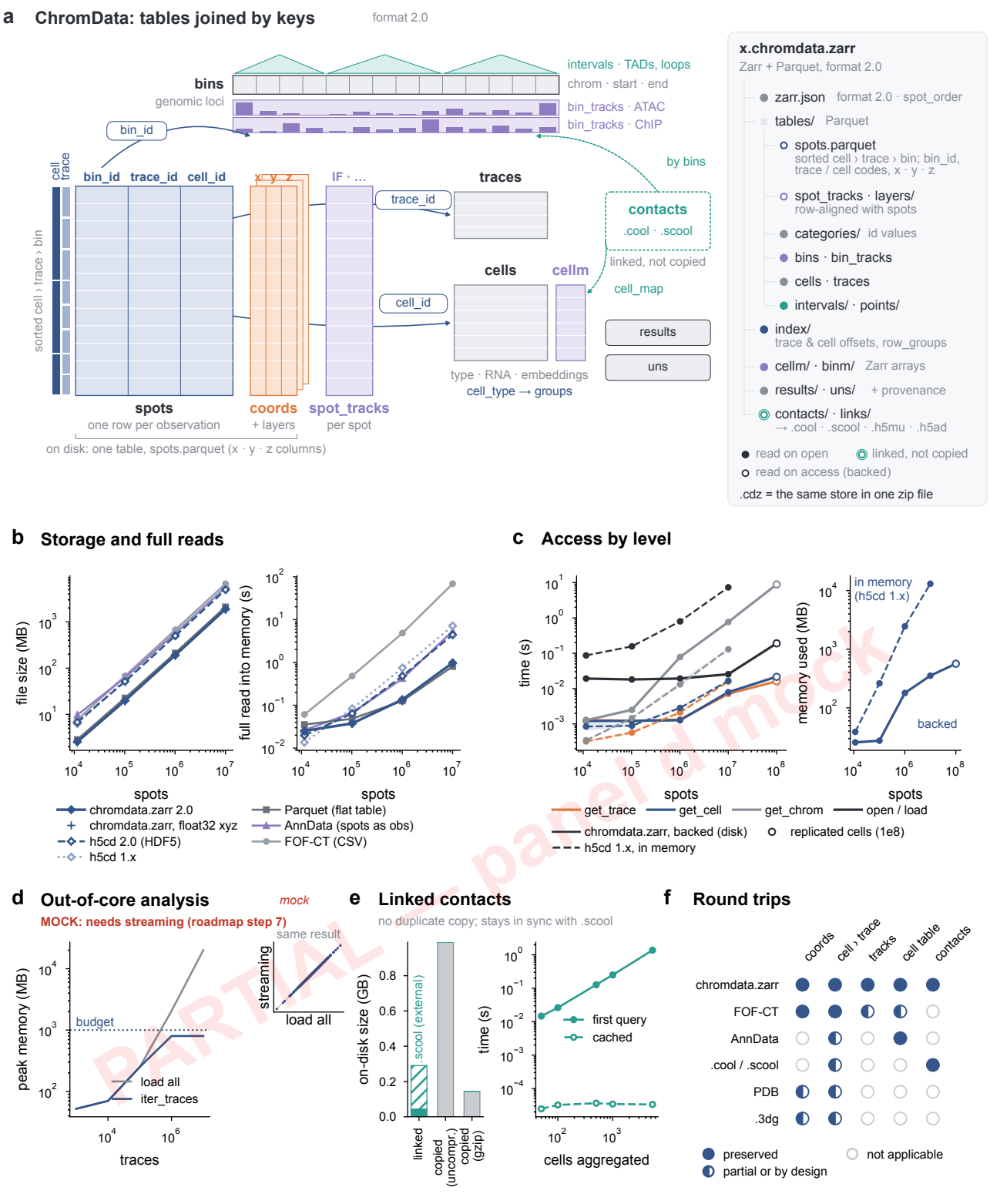

a
ChromData: tables joined by keys
format 2.0
bins
genomic loci
intervals · TADs, loops
chrom · start · end
bin_tracks · ATAC
bin_tracks · ChIP
bin_id
by bins
cell
trace
bin_id
trace_id
cell_id
x y z
IF · …
traces
trace_id
contacts
.cool · .scool
linked, not copied
cells
cellm
cell_map
cell_id
results
uns
sorted cell › trace › bin
type · RNA · embeddings
cell_type → groups
spots
one row per observation
coords
+ layers
spot_tracks
per spot
on disk: one table, spots.parquet (x · y · z columns)
x.chromdata.zarr
Zarr + Parquet, format 2.0
zarr.json format 2.0 · spot_order
tables/ Parquet
spots.parquet
sorted cell › trace › bin; bin_id, trace / cell codes, x · y · z
spot_tracks · layers/
row-aligned with spots
categories/ id values
bins · bin_tracks
cells · traces
intervals/ · points/
index/
trace & cell offsets, row_groups
cellm/ · binm/ Zarr arrays
results/ · uns/ + provenance
contacts/ · links/
→ .cool · .scool · .h5mu · .h5ad
read on open
linked, not copied
read on access (backed)
.cdz = the same store in one zip file
b
Storage and full reads
file size (MB)
full read into memory (s)
spots
spots
chromdata.zarr 2.0
chromdata.zarr, float32 xyz
h5cd 2.0 (HDF5)
h5cd 1.x
Parquet (flat table)
AnnData (spots as obs)
FOF-CT (CSV)
c
Access by level
time (s)
memory used (MB)
in memory
(h5cd 1.x)
backed
spots
spots
get_trace
get_cell
get_chrom
open / load
chromdata.zarr, backed (disk)
replicated cells (1e8)
h5cd 1.x, in memory
d
Out-of-core analysis
mock
MOCK: needs streaming (roadmap step 7)
peak memory (MB)
budget
load all
iter_traces
traces
same result
streaming
load all
e
Linked contacts
no duplicate copy; stays in sync with .scool
on-disk size (GB)
scool (external)
linked
copied (uncompr.)
copied (gzip)
time (s)
first query
cached
cells aggregated
f
Round trips
coords
cell › trace
tracks
cell table
contacts
chromdata.zarr
FOF-CT
AnnData
.cool / .scool
PDB
.3dg
preserved
not applicable
partial or by design
PARTIAL — panel d mock